Ex: 01. Prove that $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$ if $R \subseteq A \times B$ and $S \subseteq B \times C$. Ex

02. Using $\left(\bigcap_{\alpha \in \Lambda} A_\alpha\right)^c = \bigcup_{\alpha \in \Lambda} A_\alpha^c$ and $\left(A^c\right)^c = A$,

~~prove~~ deduce that $\left(\bigcup_{\alpha \in \Lambda} A_\alpha\right)^c = \bigcap_{\alpha \in \Lambda} A_\alpha^c$

Proof: $\left(\bigcap_{\alpha \in \Lambda} A_\alpha^c\right)^c = \bigcup_{\alpha \in \Lambda} (A_\alpha^c)^c = \bigcup_{\alpha \in \Lambda} A_\alpha$

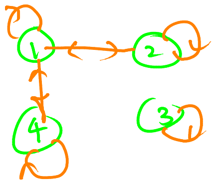
$\therefore \left[\left(\bigcap_{\alpha \in \Lambda} A_\alpha^c\right)^c\right]^c = \left(\bigcup_{\alpha \in \Lambda} A_\alpha\right)^c$

Hence, $\left(\bigcup_{\alpha \in \Lambda} A_\alpha\right)^c = \bigcap_{\alpha \in \Lambda} A_\alpha^c$ ∎

Properties of Relations

There are several properties that are used to classify relations on a set. We will introduce the most important of these here. $\mathcal{R} \subseteq A \times A$

1. **Reflexive :** A relation $\mathcal{R}$ on a set A is called *reflexive* if $(a,a) \in \mathcal{R}$ for every element $a \in A$.
   In other words, $\forall a \in A,\ (a,a) \in \mathcal{R}$

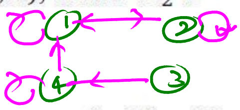
   

   Examples : Consider the following relations on $\{1,2,3,4\}$

   $\mathcal{R}_1 = \{(1,1),(1,2),(1,4),(2,1),(2,2),(3,3),(4,1),(4,4)\}$

   $\mathcal{R}_2 = \{(1,1),(1,2),(2,1),(2,2),(3,4),(4,1),(4,4)\}$ $(4,3) \notin \mathcal{R}_2$

   The relation $\mathcal{R}_1$ is reflexive because it both contain all pairs of the form (a,a), while $\mathcal{R}_2$ is not reflexive. (Since $(3,3) \notin \mathcal{R}_2$)

2. **Symmetric :** A relation $\mathcal{R}$ on a set A is called *symmetric* if $(b,a) \in \mathcal{R}$ whenever $(a,b) \in \mathcal{R}$, for all $a,b \in A$.
   In other words, $\forall a \in A, \forall b \in A,\ (a,b) \in \mathcal{R} \implies (b,a) \in \mathcal{R}$
   Examples : Consider the above example :
   $\mathcal{R}_1$ is symmetric, because in each case (b,a) belongs to the relation whenever (a,b) does.
   $\mathcal{R}_2$ is not symmetric, because there exists $(3,4) \in \mathcal{R}_2$ but $(4,3) \notin \mathcal{R}_2$.

3. **Antisymmetric :** A relation $\mathcal{R}$ on a set A is called *antisymmetric* if $(a,b) \in \mathcal{R}$ and $(b,a) \in \mathcal{R}$ then $a = b$, for all $a,b \in A$.
   In other words, $\forall a \in A, \forall b \in A,\ (a,b) \in \mathcal{R} \wedge (b,a) \in \mathcal{R} \implies a = b$ $\equiv \forall a \in A\ \forall b \in A,\ a \neq b \Rightarrow (a,b) \notin \mathcal{R} \vee (b,a) \notin \mathcal{R}$
   Examples : Consider the following relations on $\{1,2,3,4\}$
   $\mathcal{R}_3 = \{(2,1),(3,1),(3,2),(4,1),(4,2),(4,3)\}$
   $\mathcal{R}_4 = \{(1,1),(1,2),(2,1),(2,2),(3,4),(4,1),(4,4)\}$
   $\mathcal{R}_3$ is antisymmetric, because there is no pair of elements a and b with $a \neq b$ such that both (a,b) and (b,a) belong to the relation.
   * $\mathcal{R}_4$ is not antisymmetric, because there is $(1,2) \in \mathcal{R}_4$ and $(2,1) \in \mathcal{R}_4$ but $1 \neq 2$.

   Ex: Let R be a relation on a set A.

   Prove that R is antisymmetric if and only if $R \cap R^{-1} \subseteq I_A$.

   $R_3^{-1} = \{(1,2),(1,3),(2,3),(1,4),(2,4),(3,4)\}$ $R_3 \cap R_3^{-1} = \{\ \} = \emptyset \subseteq I_A$ $\therefore$ R_3 is antisymmetric

4. **Transitive :** A relation $\mathcal{R}$ on a set A is called *transitive* if whenever $(a,b) \in \mathcal{R}$ and $(b,c) \in \mathcal{R}$, then $(a,c) \in \mathcal{R}$, for all $a,b,c \in A$.
   In other words, "$\forall a,b,c \in A,\ (a,b) \in \mathcal{R} \wedge (b,c) \in \mathcal{R} \implies (a,c) \in \mathcal{R}$"
   Examples : $\mathcal{R}_3$ is transitive because if (a,b) and (b,c) belong to $\mathcal{R}_3$, then (a,c) also does.
   $\mathcal{R}_1$ is not transitive because $(4,1)$ and $(1,2)$ belong to $\mathcal{R}_1$, but $(4,2)$ does not.

Note:

Definition: Suppose $\mathcal{R}$ is a relation on set A. Then:

(i) $\mathcal{R}$ is said to be *reflexive* if $\forall x \in A, x\mathcal{R}x$

(ii) $\mathcal{R}$ is said to be *symmetric* if $\forall x, y \in A, x\mathcal{R}y \Rightarrow y\mathcal{R}x$

(iii) $\mathcal{R}$ is said to be *transitive* if $\forall x, y, z \in A, (x\mathcal{R}y \wedge y\mathcal{R}z) \Rightarrow x\mathcal{R}z$

Definition

A relation R on A is an **equivalence relation** if R is reflexive, symmetric, and transitive.

Example1: The relation "=" is an equivalence relation on IR.

Example2: Let σ is on $\mathbb{Z}$ and $\forall x, y \in \mathbb{Z}, x\sigma y$ if and only if $3\,|(x-y)$ is an equivalence relation on $\mathbb{Z}$.

$\sigma = \{(x,y) \mid x,y \in \mathbb{Z},\ x\sigma y\} = \{(x,y) \mid x,y \in \mathbb{Z},\ 3 \mid x-y\}$

Exercise : Determine whether the relation $\mathcal{R}$ is reflexive, symmetric, antisymmetric, and/or transitive.

$$\forall x, y \in \mathbb{R}, \quad x\mathcal{R}y \iff x^2 - y^2 = x - y$$

$\mathcal{R} = \{(x,y) \mid x,y \in \mathbb{R},\ x^2 - y^2 = x - y\}$.

Note:

The symbol $\sim$ is almost universally used for an abstract equivalence relation. It can be read as 'related to.'

Example. For all $x, y \in \mathbb{Z}$, let $x \sim y \iff x - y$ is even. We claim that $\sim$ is an equivalence relation on $\mathbb{Z}$.

Reflexivity $\forall x \in \mathbb{Z}$, $x - x = 0$ is even, hence $x \sim x$.

Symmetry $\forall x, y \in \mathbb{Z}$, $x \sim y \implies x - y$ is even $\implies y - x$ is even $\implies y \sim x$.

Transitivity $\forall x, y, z \in \mathbb{Z}$, if $x \sim y$ and $y \sim z$, then $x - y$ and $y - z$ are even. But the sum of two even numbers is even, hence $x - z = (x - y) + (y - z)$ is even, and so $x \sim z$.

Equivalence Classes :

Let $\mathcal{R}$ be an equivalence relation on A. Let $a \in A$,

the equivalence class of a denoted by $\mathcal{C}(a)$ is defined as the set of all those point of A which are related to a under the relation $\mathcal{R}$. ie $\mathcal{C}(a) = \{x \in A \mid x\mathcal{R}a\}$

- Let $\mathcal{R}$ be an equivalence relation on A and $a, b \in A$, where $a\mathcal{R}b$, then a and b have the same equivalence class.

Note:

Definition. Let $\sim$ be an equivalence relation on A. The *equivalence class* of x is the set

$$[x] = \{y \in \mathrm{A} : y \sim x\}.$$

Examples.

01. $[0] = \{y \in \mathbb{Z} : y \sim 0\} = \{y \in \mathbb{Z} : y \text{ is even}\}$ is the set of even numbers. Note that $[0]$ is also equal to $[2]$, $[4]$, $[6]$, etc.
The other equivalence class is $[1] = \{y \in \mathbb{Z} : y - 1 \text{ is even}\}$, which is the set of odd numbers.

02. $[0] = \{y \in \mathbb{Z} : y \sim 0\} = \{y \in \mathbb{Z} : 3 \mid y - 0\}$

$[5] = \{y \in \mathbb{Z} : y \sim 5\} = \{y \in \mathbb{Z} : 3 \mid y - 5\}$

Note that $x \sim y$ if and only if $3 \mid x - y$.

$[7] = ?$

$\mathbb{Z} = [0] \cup [1] \cup [2]$

$[0] \cap [1] = \phi$ $\quad [0] \cap [2] = \phi$

$[1] \cap [2] = \phi$

End of D_{21}